TATHYA DUBEY

AI & LLM Systems Engineer | B.Tech CSE, 2027
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SUMMARY

CS undergraduate (CGPA 8.71) who builds AI systems end to end: multi-agent orchestration and retrieval pipelines, plus the infrastructure that makes them dependable in production, including rate limiting, model failover, sandboxing, and observability. Seeking AI / LLM systems engineering roles (remote-friendly).

TECHNICAL SKILLS

AI & Machine Learning: LangGraph, RAG, Multi-Agent Systems, Scikit-Learn, faster-whisper, Kokoro TTS, Gemini & Anthropic APIs, Ollama, Cross-Encoder Reranking, Semantic Caching, PromptOps.
Vector & Data Stores: PostgreSQL (pgvector), Supabase, Redis (Lua / Pub-Sub), SQLite, Pandas, NumPy.
Systems & Infrastructure: Python, FastAPI, Celery, Kafka/RabbitMQ, S3/MinIO, Docker (Sandboxing & Compose), Linux systemd, D-Bus, OpenTelemetry (Jaeger), AST Analysis, Semgrep, C, TypeScript, Next.js.

EXPERIENCE

THEKULTHOUSE | Co-Founder & Technical Lead **Jun 2026 – Present**
- Spearheaded architecture and delivery of bespoke SaaS applications, AI-integrated workflows, and high-conversion web infrastructure for commercial clients.
- Built end-to-end full-stack solutions: scalable APIs, automated booking/data pipelines, low-latency UIs.

HostingSpell | Cybersecurity Intern **Nov 2025 – Dec 2025**
- Executed vulnerability assessments, OWASP Top 10 audits, and secure-coding remediation of injection and auth flaws across production web applications and hosting infrastructure.

PROJECTS

DevPulse | Autonomous Multi-Agent Code Reviewer | *LangGraph, FastAPI, pgvector, Redis, Celery, OTel*
- Architected a LangGraph code-auditing graph that clones repos, retrieves security/style rules from a pgvector knowledge base, and runs parallel Security and Architecture review agents.
- Built a fault-tolerant LLM gateway with a Redis Lua atomic token-bucket limiter (tested at 200 concurrent threads) and deterministic failover: Anthropic → Gemini → Ollama → Python AST/Semgrep.
- Added a line-preserving PII/secret sanitizer (Luhn-checked cards, API keys, JWTs), zero-network Docker sandboxing, versioned PromptOps, and OTel GenAI cost/token tracing.

SMTEKE | Distributed Multi-Tenant Enterprise Engine | *Python, Kafka/RabbitMQ, Redis, S3/MinIO, Docker*
- Designed an async B2B backend ingesting 500+ page documents via S3/MinIO, Kafka/RabbitMQ queues, and horizontally scaled Docker workers with Snowflake ID tracing.
- Prevented cache stampedes with probabilistic early expiration in Redis, isolated noisy-neighbor tenants with sliding-window rate limits, and served hybrid BM25 + vector search with cross-encoder reranking per tenant.

Nova | Local Linux Voice AI Assistant | *Python, systemd, D-Bus, faster-whisper, Kokoro TTS*
- Orchestrated a fully offline voice assistant across 7 systemd daemons over D-Bus and POSIX IPC.
- Routed faster-whisper transcriptions through intent classification to system commands, a local RAG memory, or Ollama LLMs, replying through Kokoro TTS.

Production ML & GenAI Suite | *Python, Scikit-Learn, LangGraph, Gemini API, Pytest*
- Trained leak-free fraud/churn pipelines for severe class imbalance (anomaly detection, PR-AUC, calibration, cost-based thresholds) and a citation-enforced RAG system with an automated evaluation harness.

EDUCATION & CERTIFICATIONS

Indus Institute of Technology and Engineering **Expected 2027**
B.Tech in Computer Science & Engineering | **CGPA: 8.71 / 10.0**
- AWS Academy Graduate: Machine Learning Foundations (2026) – credly.com/go/V436gx4Q
- AWS Academy Graduate: ML for Natural Language Processing (2026) – credly.com/go/l6gon3el